

TOPOLOGICAL FIELD THEORY AND TWO-DIMENSIONAL ABELIAN-HIGGS INSTANTONS

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We derive a topological field theory which describes the two-dimensional abelian-Higgs instanton (vortex) moduli space. In contrast to the three-dimensional monopole case here one explicitly breaks the gauge symmetry. Also a comparison is made with other supersymmetric models.

1. Introduction

A quantum field theoretic method of arriving at instanton moduli space has recently been given by Witten after initial developments by Atiyah [1]. One point of view is that the action of a topological field theory is the BRST quantization of the Langevin equation that one may associate with the instanton defining equation [2,3], though there are other ways of getting to the same result [4–7].

It is possible to construct a three-dimensional topological field theory which gives information about the moduli space of three-dimensional instantons of the Georgi–Glashow $SO(3)$ model [3,8]. These instantons were first considered by Polyakov [9]. Unfortunately the information gleaned in this way is not complete. The topological field theory in three dimensions, only deals with the Prasad–Sommerfield limit [10] as there is no ϕ^4 Higgs coupling. Indeed the three-dimensional model is obtained from Witten action by dimensional reduction (all fields x^4 independent and $A_4 \rightarrow$ Higgs field) and so one is dealing with static solutions (in the four-dimensional space) i.e. monopoles in the Prasad–Sommerfield limit. This situation is somewhat disappointing as it seems impossible to introduce a Higgs coupling while keeping the topological nature of the theory; consequently only

a portion of the instanton moduli space is being considered.

The reduction may be made one more time. Indeed in ref. [3] one of us and Birmingham and Rakowski wrote down the starting action that one gets in this way. The reduction has all fields x_3, x_4 independent while A_3, A_4 become the components of a complex Higgs field. The moduli space that this entails is that considered by Hitchin [11] (in giving a treatment of a parametrization of the Teichmüller space of compact Riemann surfaces). However, in two dimensions there is another set of important $U(1)$ instantons associated with the abelian-Higgs model [12] (these instantons are also vortex (time independent, axially symmetric) solutions in the four-dimensional sense [13]). To get to the $SO(2)$ model by reduction requires one to take $A_\mu^1 = A_\mu^2 = 0, A_3^a = A_4^a = 0$ ($\mu = 1, 2; a = 1, 2, 3$). Then A_μ^3 is in the adjoint and A_4^a is in the fundamental of $SO(2)$. The problem that one encounters now is that again there are no ϕ^4 or ϕ^2 couplings and one cannot get in this way to the Bogomol'nyi equations [14].

In this letter we will show that, happily, it is possible to introduce these couplings. Indeed a straightforward BRST quantization a la refs. [2–4] of the two-dimensional Bogomol'nyi equations will be presented. Relevant topological invariants on the moduli space of the theory arise as the c_n term of the Chern polynomials introduced in refs. [3,4] and also as functions of the Higgs fields and their ghosts.

Because of the ability to get around the standard

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spin-statistics theorem in two dimensions we will be able to write the BRST ghosts as bonafide spinors. With the action written in this form, there is a starting resemblance to the four-dimensional supersymmetric abelian-Higgs model [15,16]. The correspondence is even more tantalizing when one notices that the Higgs coupling λ is related to the gauge coupling e as $\lambda = e$ in both the four-dimensional and two-dimensional theories [14–17]. In $d=4$, $\lambda=e$ is required in the supersymmetric theory in order that all fields have the same mass ($m=e\phi_0$); it also becomes at this point a parity-conserving theory.

2. Topological Bogomol'nyi action

The starting action is taken to be

$$S_c = a_1 \int L^2 + \int K_\alpha^a K_\beta^b S_{ab} g^{\alpha\beta}, \quad (1)$$

with

$$L = G - \frac{1}{2} \epsilon^{\alpha\beta} F_{\alpha\beta} - a_2 (a_3^2 - \phi^2),$$

$$K_\alpha^a = G_\alpha^a - \frac{1}{2} (D_\alpha \phi^a + \epsilon_\alpha^\beta \epsilon^a_b D_\beta \phi^b), \quad (2)$$

where G and G_α^a are random fields [2–4] and K_α^a and G_α^a are self-dual, i.e. $K_\alpha^a = \epsilon_\alpha^\beta \epsilon^a_b K_\beta^b$ (and likewise for G_α^a). Our notation is that

$$D_\alpha \phi^a = \partial_\alpha \phi^a - i \epsilon_\alpha^\beta A_\beta \phi^a, \quad (3)$$

and $\sqrt{g}$ factors have been suppressed throughout.

Some remarks concerning the constants that appear in (1) are in order. By scaling ϕ one may set $a_3^2 = 1$, while scaling G_α^a by the same amount has the effect of altering a_1 and a_2 . Now the altered a_2 , a_2' may be set to 1 via the transformations

$$A_\mu(x) \rightarrow \alpha A_\mu(\alpha x), \quad \phi(x) \rightarrow \phi(\alpha x),$$

$$x \rightarrow \alpha x, \quad \alpha = (a_2')^{1/2}.$$

So we are left with the free parameter a_1 . This freedom is not of any concern as the minimum of the action with $a_2 = 1 = a_3$ occurs at (with the random field at zero)

$$\frac{1}{2} \epsilon^{\alpha\beta} F_{\alpha\beta} = 1 - \phi^2,$$

$$D_\alpha \phi^a + \epsilon_\alpha^\beta \epsilon^a_b D_\beta \phi^b = 0. \quad (4)$$

These are the Bogomol'nyi equations. When a_1 is chosen to be $\frac{1}{2}$ (again with $G = G_\alpha^a = 0$) so that (1)

coincides with the abelian-Higgs model, one sees that the Higgs coupling $\frac{1}{4} \lambda (\phi^2 - 1)^2$ is given by $\lambda = e^{\#1}$. This point in parameter space divides type-I and type-II superconductors and represents a point of no interaction between the vortices. Coupled with this it must be recalled that the two-dimensional stress-energy tensor of the (still to be derived) topological field theory is a BRST commutator, and hence has zero expectation value. There are no interacting states of any kind in this model.

The quantization of (1) proceeds along the lines spelt out in refs. [3,5], in the spirit of Batalin and Vilkovisky [18].

The symmetries to be dealt with are

$$\delta A_\mu = -\partial_\mu \omega + \epsilon_\mu,$$

$$\delta \phi^a = \lambda^a - i \epsilon_\alpha^\beta \omega \phi^b,$$

$$\delta G = \epsilon^{\alpha\beta} \partial_\alpha \epsilon_\beta - 2 \lambda^a \phi_a,$$

$$\delta G_\alpha^a = \frac{1}{2} (D_\alpha \lambda^a + \epsilon_\alpha^\beta \epsilon^a_b D_\beta \lambda^b)$$

$$- \frac{1}{2} i (\epsilon_\alpha^\beta \epsilon_\beta^\gamma \phi^\gamma + \epsilon_\alpha^\beta \epsilon_\beta^\gamma \phi^\gamma) - i \epsilon_\alpha^\beta \omega G_\beta^a. \quad (5)$$

The only subtlety is that there is an on-shell zero-mode when $\omega = A$, $\epsilon_\mu = \partial_\mu A$, $\lambda^a = i \epsilon_\alpha^\beta A \phi^b$, so that the variations in (5) all vanish except that for G_α^a which reduces to

$$\delta G_\alpha^a = -i \epsilon_\alpha^\beta A K_\beta^a, \quad (6)$$

and which vanishes only when the K_α^a equation of motion is employed. To each of the symmetries above we assign a ghost field:

$$\epsilon_\alpha \rightarrow \psi_\alpha, \quad \omega \rightarrow c, \quad \lambda^a \rightarrow \rho^a, \quad A \rightarrow \eta. \quad (7)$$

The ghost numbers of the minimal set (A_μ , ϕ^a , G , G_α^a , ψ_μ , c , ρ^a , η) are (0, 0, 0, 0, 1, 1, 1, 2) and we introduce antifields ($A^{\mu*}$, ϕ_a^* , G^* , $G_{\alpha*}^a$, $\psi^{\mu*}$, c^* , ρ_a^* , η^*) with ghost numbers (-1, -1, -1, -1, -2, -2, -2, -3). The minimal solution to the master equation [18] we find is

$$\begin{aligned} S_{\min} = & S_c + A^{\mu*} \delta A_\mu + \phi_a^* \delta \phi^a \\ & + G^* \delta G + G_{\alpha*}^a \delta G_\alpha^a + c^* \eta + \psi^{\alpha*} \partial_\alpha \eta \\ & + \rho_a^* (i \epsilon_\alpha^\beta \phi^b \eta + i \epsilon_\alpha^\beta \rho^b) \\ & + G_{\beta*}^b G_{\alpha*}^a \frac{1}{4} i \epsilon^{ab} \delta_{\alpha\beta} \eta, \end{aligned} \quad (8)$$

where the variations in (8) are as in (6) except with

^{#1} As in eq. (3) we take $e=1$ for simplicity.

the identifications (7). The action (8) is supplemented with new fields so that

$$S = S_{\min} + \chi_{\alpha}^{*a} (d^{\alpha}{}_a + c \epsilon_a{}^b d^{\alpha}{}_b) + \omega^{*} \Omega + b^{*} d - i d^{\alpha}{}_a \epsilon_a{}^b d_b{}^{\alpha} c. \quad (9)$$

The gauge fermion which enforces the gauge conditions

$$0 = G = G_{\alpha}^a = \partial_{\mu} A^{\mu} = \partial^{\alpha} \psi_{\alpha}$$

is

$$\psi = \chi_a^{\alpha} G_{\alpha}^a + b \partial^{\alpha} A_{\alpha} + \omega G + \lambda \partial^{\alpha} \psi_{\alpha}. \quad (10)$$

The quantum action is obtained on restricting the antfields to lie on the surface $\Phi^{*} = \partial \psi / \partial \Phi$. Taking this into account and setting $G = G_{\alpha}^a = 0$ we arrive at

$$S = [\frac{1}{2} \epsilon^{\alpha\beta} F_{\alpha\beta} + (1 - \phi^2)] \Omega + \frac{1}{2} D_{\alpha} \phi^a d^{\alpha}{}_a + d^{\alpha}{}_a d^a{}_{\alpha} + (1/a_1) \Omega^2 + b(-\partial^2 c + \partial^{\mu} \psi_{\mu}) + \omega(\epsilon^{\alpha\beta} \partial_{\beta} \psi_{\alpha} - 2a_2 \rho^a \phi_a) + \chi_a^{\alpha} D_{\alpha} \rho^a - i \chi_a^{\alpha} \epsilon_a{}^b \psi_{\alpha} \phi^b + \lambda \partial^2 \eta + d \partial_{\mu} A^{\mu} - \sigma \partial^{\mu} \psi_{\mu} + \chi_a^{\alpha} \chi_b^{\beta} i \epsilon^{ab} g_{\alpha\beta} \eta. \quad (11)$$

with gauge transformations

$$\begin{aligned} \delta A_{\mu} &= \epsilon(\partial_{\mu} c - \psi_{\mu}), \\ \delta c &= \epsilon \eta, \quad \delta \omega = \epsilon \Omega, \\ \delta \phi^a &= -\epsilon(\rho^a - i c \epsilon^a{}_b \phi^b), \\ \delta \eta &= 0, \quad \delta \Omega = 0, \\ \delta \chi_{\alpha}^a &= \epsilon(d^a{}_{\alpha} + i \epsilon^a{}_b c \chi^b{}_{\alpha}), \\ \delta \psi_{\mu} &= \epsilon \partial_{\mu} \eta, \quad \delta \lambda = \epsilon \sigma, \\ \delta \rho^a &= \epsilon(i \epsilon^a{}_b \phi^b \eta + i c \epsilon^a{}_b \rho^b), \\ \delta d &= 0, \quad \delta b = \epsilon d, \quad \delta \sigma = 0, \\ \delta d^a{}_{\alpha} &= i \epsilon \epsilon^a{}_b c d^b{}_{\alpha} - i \epsilon \epsilon^a{}_b \eta \chi^b{}_{\alpha}. \end{aligned} \quad (12)$$

The action and transformation rules as given may be derived on elimination of the random fields from simple but lengthy arguments presented in refs. [2,3]. One may check that (12) is a symmetry of (11). Indeed given the transformation rules (12) we may show that S is a BRST commutator,

$$S = \{Q, \omega(\frac{1}{2} \epsilon^{\alpha\beta} F_{\alpha\beta} + (1 - \phi^2) + (1/a_1) \Omega) + b \partial_{\mu} A^{\mu} + \lambda \partial^{\mu} \psi_{\mu} + \chi_{\alpha}^a (d^a{}_{\alpha} + D_{\alpha} \phi^a)\}, \quad (13)$$

where $\epsilon Q = \delta$ and $Q^2 = 0$. The usual arguments now apply [1] so that one knows that the stress-energy tensor is also a BRST commutator.

The theory that follows from (11) is then the topological field theory of interest; however, it is not unique in two respects. Firstly, one may add to (11) the BRST variation of anything providing this does not ruin renormalizability, the damping of the action and so on. The second reason why it is not unique has to do with the gauge transformations for the vector potential A_{μ} in (5). The transformation is as is given in all previous works [1-8], however in two dimensions, we could equally well have chosen the Hodge decomposition for the transformation

$$\delta A = d\omega + *d*\epsilon + \gamma, \quad \gamma \text{ harmonic}.$$

In this case there is no zero-mode associated with the symmetry. The transformations one finds are as in (12) except $\psi_{\mu} \rightarrow \epsilon_{\mu}{}^{\nu} \partial_{\nu} \psi$ and η and λ are set to zero. An appropriate action is (13) with the above restrictions. Because of the lack of a zero-mode one could forgo the Batalin-Vilkovisky machinery and apply the time-honored Faddeev-Popov construction directly. The question of zero-modes has been pushed into the handling of γ , which is straightforward. However, for our present purposes the action and transformations (11) and (12) are more suitable.

3. Comparison with other supersymmetric models

Our aim in this section is to rewrite the ghost terms in (11) as true fermions. This can be done in two dimensions because of the triviality of the Lorentz group (and consequently the weakness of the spin-statistics theorem)^{#2}. Before doing this we add to (11) the term

$$\begin{aligned} \{Q, \epsilon_{ab} \phi^a \rho^b \lambda\} &= \epsilon_{ab} (\rho^a - i c \epsilon^a{}_d \phi^d) \rho^b \lambda \\ &\quad - \epsilon_{ab} \phi^a (i \epsilon^a{}_d \phi^d \eta + i c \epsilon^a{}_d \rho^d) \lambda \\ &\quad + \epsilon_{ab} \phi^a \rho^b \sigma, \end{aligned} \quad (14)$$

without spoiling the symmetry. The ψ zero-mode measures the dimension of the moduli space around a given instanton and as is known from previous work

^{#2} An example of this may be found in ref. [2].

[9,12] there is a "natural" gauge condition on the zero modes given by

$$\partial^\mu \psi_\mu + \epsilon_{ab} \phi^a \rho^b = 0.$$

The term we have added leads to this condition in the weak-coupling limit. Now we make the following identifications:

$$\psi = \begin{pmatrix} \sigma + i\omega \\ \psi_- \end{pmatrix}, \quad \bar{\psi} = -(\sigma - i\omega, \psi_+),$$

$$\chi = \frac{1}{\sqrt{2}} \begin{pmatrix} \rho^- \\ \chi_-^+ \end{pmatrix}, \quad \bar{\chi} = \frac{1}{\sqrt{2}} (\rho^+, \chi_-^+),$$

$$A = \frac{1}{\sqrt{2}} (\lambda + \eta), \quad B = \frac{1}{\sqrt{2}} (\lambda - \eta).$$

In these variables (11) together with (14) becomes (ignoring the normal gauge potential Faddeev-Popov terms)

$$\begin{aligned} S = & S_c + \bar{\chi} \not{D} \chi + \bar{\psi} \not{D} \psi \\ & + \sqrt{2} e (\bar{\chi} \psi \phi^- + \bar{\psi} \chi \phi^+) + 2\sqrt{2} i e \bar{\chi} (A + \gamma_5 B) \chi \\ & - e^2 |\phi|^2 (A^2 - B^2) + \frac{1}{2} A \partial^2 A - \frac{1}{2} B \partial^2 B. \end{aligned} \quad (15)$$

In form this bears a striking resemblance to the U(1) parity-conserving supersymmetric abelian-Higgs model introduced by Salam and Strathdee [15]. The interesting point is that these authors and Fayet [16] also find the restriction $\lambda = e!$ But here we have what would be a (twisted) $N=2$ supersymmetry rather than $N=1$ so it is not apparent how (15) might arise from the reduction (which is tempting) of the higher-dimensional models. Such a correspondence would finally explain the reason why the Higgs coupling is so constrained in two and four dimensions. We believe (15) goes some way in this direction. It is also in this regard that (15) differs from the work of Di Vecchia and Ferrara [19] as their two-dimensional supersymmetric abelian-Higgs Model is an $N=1$ model. One can also trivially construct a topological supersymmetric field theory based on the supersymmetric instanton equations of this paper, as done in ref. [20]. In fact the resulting two-dimensional model will be the dimensionally reduced version of that in ref. [20]. It would be interesting to see if one could create a standard $N=2$ model that corresponds with (15).

4. Topological invariants

The first Chern term $\int F$ is the obvious topological invariant available in this theory. Associated with it are the terms η and ψ_μ , so that

$$0 = \{Q, \eta\}, \quad d\eta = \{Q, \psi\},$$

$$d\psi = \{Q, F\}, \quad dF = 0$$

on the moduli space (we mean it quotiented with the gauge group). To build more invariants we note that the three fields η , ψ and F form the basic blocks of the Chern polynomials introduced in refs. [3,4]. Indeed the sequence above arises as the first Chern class. Other invariants are obtained by considering the n th Chern class,

$$0 = \{Q, \eta^n\},$$

$$d\eta^n = \{Q, n\psi\eta^{n-1}\},$$

$$d(n\psi\eta^{n-1}) = \{Q, -nF\eta^{n-1} + \frac{1}{2}n(n-1)\psi^2\eta^{n-2}\},$$

$$d(-nF\eta^{n-1} + \frac{1}{2}n(n-1)\psi^2\eta^{n-2}) = 0.$$

How to evaluate these invariants on the moduli space is given in some detail in ref. [1]. These terms all have to do with the vector potential and consequent ghosts. We may also build invariants out of the Higgs field and its ghosts. For this purpose we note that with

$$W_2 = \frac{1}{2} \epsilon_{ab} d\phi^a d\phi^b, \quad W_1 = \epsilon_{ab} \rho^a d\phi^b,$$

$$dW_1 = \{Q, W_2\}, \quad dW_2 = 0.$$

W_1 and W_2 may also be obtained by the dimensional reduction of W_3 and W_4 of the four-dimensional theory [1]. Notice that $\int W_2$ measures the same winding number as $\int F$ as it evaluates the phase of the Higgs field around the circle at infinity. W_1 is not an invariant as $\{Q, W_1\} \neq dG$ for some G .

In summary we have shown that a topological field theory of abelian-Higgs instantons may be easily constructed. It does not arise as the trivial reduction of Witten's model in four dimensions, though its resulting form is very much like that of the supersymmetric abelian-Higgs model in four dimensions. The similarity is really very striking and it seems worthwhile to determine whether the two theories are related by some reduction.

There is an alternative cohomological construction that is available, based on equivariant cohomology

[21] which is close in spirit to the work presented in ref. [7]. A similar analysis is also possible in the model we have discussed.

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Note added

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